

Akshay Patil

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EDUCATION

Georgia Institute of Technology

August 2026 – May 2029

Bachelor of Science in Aerospace Engineering

School of Science and Engineering at Yvonne A. Ewell Townview Center – Dallas, TX

August 2022 – May 2026

3.97 Unweighted GPA | Rank 2 | 21 Advanced Placement Classes | 1600 SAT | National Merit Scholar | NASA High School Aerospace Scholar

Relevant Coursework: AP Physics C Mechanics, AP Physics C E&M, AP Calculus BC, AP Chemistry, AP Computer Science A, AP Research

EXPERIENCE

Hardpoint Robotics | Co-Founder

June 2026 – Present

- Designing and building NDAA-compliant development UAVs; FAA Part 107 Remote Pilot certified
- Designing airframes in CAD, 3D printing CF-Nylon parts, and working with vendors to manufacture the carbon-fiber structure

SWYFT Robotics | Engineering Design Intern

May 2025 – May 2026

- Designed high quality and durable sprockets, gears, and pulleys for use in FIRST Tech Challenge and Robotics Challenge competitions
- Designed modular and versatile robot cart for FIRST Robotics Competition teams with a detachable programming station
- Learned how to manufacture high quality parts for low prices using extrusion and fiber laser cutting techniques

CLUBS & ACTIVITIES

FIRST TECH CHALLENGE | Captain and Lead Designer of Team 7172 (ftc7172.org)

September 2021 – May 2026

- Team is currently internationally ranked 3rd and was part of the Finalist Alliance at the 2026 World Championship
- Designed mechanisms using various mechanical concepts including linkages, counter-springing, pulley systems, geartrains, etc.
- Used industrial arm mechanics to design and build robotic arms with 6 degrees of freedom
- 24 Competition Wins (2021-2026) – plus Inspire, Think, Connect, Innovate, Design, Motivate, and Control awards

Beaver Works Summer Institute at MIT | Participant

June 2024 – August 2024

- Built a Synthetic Aperture Radar system to image obscured objects using a PulsOn 440 radar attached to a DJI F550 UAV
- Used NumPy vectorized functions to write a back-projection algorithm to generate images from radar scans
- Used Matplotlib to create Range-Time-Intensity plots for radar scans and display and modify back-projection images
- Used Pandas to read and filter data from an IR motion-capture system

ROCKETRY CLUB | President

March 2023 – May 2026

- Designed, built, flew, and often crashed fully 3D-printed model rockets
- Learned alongside my team to use industry tools including simulators and design calculators
- Faced and mostly overcame issues of rockets failing to launch, parachute failure, and rockets going up and not coming back down

ACADEMIC DECATHLON | Co-Captain

September 2023 – May 2026

- Team placed 2nd at State; earned gold in mathematics, bronze in economics and literature
- Co-led preparation across ten subject areas, mastering 1000+ pages of material in 6 months

PROJECTS

Autonomous Robotics – Vision & Navigation

May 2025 – Present

- Building a UAV "ball boy" for pickleball practice; integrated Jetson Orin Nano + Arduino with RGB-D and global shutter cameras, HSV ball detection, AprilTag localization (benchmarked vs 3 alternatives), and Euler-integration trajectory prediction
- Building a Jetson-based autonomous robot that maps a house and counts windows, using the ROS 2 Nav2 stack for frontier-based exploration and path planning with RTAB-Map visual SLAM for persistent cross-session maps

SKILLS

- CAD – Autodesk Fusion 360, Onshape, SolidWorks, Certified in Autodesk Inventor, 3D Printing
- Programming – Python (NumPy, Matplotlib, Pandas, TensorFlow), JavaScript, HTML, CSS, SQL
- Embedded Systems – NVIDIA Jetson, Raspberry Pi, REV Control Hub, Arduino
- Software Tools – JupyterLab, TensorBoard, Anaconda, Android Studio, Git, Claude Code, MS Office, G Suite
- Problem Solving, Leadership, Teamwork, Critical Thinking